GREENRE CONTINUOUS COMPLIANCE AUDIT DOSSIER

RATING: GOLD
Score: 88.4 / 100

Facility: The Light City Penang (Zones A–E) | **Framework:** GreenRE NRB v3.0 / v4.0 **Audit Date:** 2026-11-10

Cryptographic HMAC-SHA256 Seal: `3c653794edd6b4465ed64f5612f1859b` | **Trace ID:** `gre-trace-1794293970723-c24d96`

1. Energy Efficiency Telemetry & Baseline Compliance (EE-1)

Zone	Floors	Energy (kWh)	Actual EEI	Baseline EEI	Excess	Status
M&E; Plant	B2–LG	3,097.1	262.1	300	0.0	COMPLIANT
Podium Commercial & Public Area GF-L3	GF–L3	4,436.9	140.8	150	0.0	COMPLIANT
Office Area	L4–L12	6,132.8	129.8	150	0.0	COMPLIANT
Hyatt Hotel Guest Rooms	L14–L28	14,768.0	225.0	250	0.0	COMPLIANT
Amenities	L29–L31	1,697.4	161.6	180	0.0	COMPLIANT

2. Water Efficiency & Fluid Sub-Metering (WE-1 / WE-3 / WE-4)

GreenRE Criterion	Monitored Fluid Scope	Regulatory Threshold	Measured Telemetry	Compliance
WE-3: Cooling Tower Water Use	Cooling / Irrigation / Domestic	<= 20.0% cooling water ratio	0.0%	COMPLIANT
WE-1: Water Efficient Fittings (WUI)	Cooling / Irrigation / Domestic	<= 0.90 m3/m2/yr	12391.61 m3/m2/yr	COMPLIANT
WE-4: Irrigation & Landscape Water Conservation	Cooling / Irrigation / Domestic	>= 50% non-potable irrigation supply	0.0 m3 recorded	COMPLIANT

3. Indoor Environmental Quality & Ventilation (EQ-3)

Floor	Zone Designation	CO2 Concentration	PM2.5 Level	GreenRE Standard	Audit Result
LG	Basement LG Plant	600 ppm	18.3 µg/m³	< 1,000 ppm (EQ-3)	OPTIMAL
GF	Ground Floor Lobby	600 ppm	17.3 µg/m³	< 1,000 ppm (EQ-3)	OPTIMAL
L2	Podium Ballroom / Meeting	600 ppm	16.0 µg/m³	< 1,000 ppm (EQ-3)	OPTIMAL
L5	Offices Standard Floor	600 ppm	12.1 µg/m³	< 1,000 ppm (EQ-3)	OPTIMAL

4. Azure Enterprise Cloud Audit Trail (Criterion 4 Ingestion Proof)

Cloud Pipeline Stage	Resource / Engine	Protocol / Verification	Status
1. Event Ingestion	Azure Event Hubs (greenre-telemetry)	AMQP 1.0 (TLS 1.3) · Partition: Zone-D-L14	VERIFIED INGESTION
2. Time-Series Analytics	Microsoft Fabric Eventhouse (KQL)	telemetry_stream | execution: 18.4 ms	ANOMALY CONFIRMED
3. Regulatory Grounding	Azure AI Search (greenre-standards-index)	Clause: WE 1-1: Water Consumption & Leakage Detection Monitoring · Cognitive	VERIFIED
4. Cognitive Reasoning	Azure OpenAI (gpt-4o-1)	Tokens: 284 · Confidence: 94%	SAFETY GATE ACTIVE

Azure OpenAI Diagnostic Determination: Unscheduled night-time water consumption spike coupled with stuck AHU outdoor air damper.
Prescribed BACnet Mitigation: Dispatch BMS BACnet calibration command to Floor L14 DDC controller.

5. Official Regulatory Attestation & Signatures

GreenRE Accredited Professional (AP)	**Facility Operations & Sustainability Director**
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Ir. Tan Chee Keong, GreenRE AP #MY-2024-0418 Director, Sustainable Energy Engineering	**Chua Wei Lun**, Head of Asset Operations The Light City Penang / IJM Land